

Two variables after the kill:

$s(\gamma_1, \text{parity})$ dies on χ_{-23}

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Abstract

The published four-variable map is dead on χ_{-23} (−29%). The candidate that remains after dropping the gap and D is $s = c\gamma_1^\alpha \rho^{[\text{odd}]}$. Fitted on the thirteen stationary rows of the depth-note table it predicts $\hat{s}(\chi_{-23}) = 1.06$ against 0.54 measured: +97%. The two-variable map dies harder than the four-variable one. The guilty axis is parity: χ_{-23} is odd and drills like even. A refit $\gamma_1 + \text{gap} + D$ *without* parity lands at 0.45 against 0.54 (−16%). That is not a new map — it is the diagnosis of the kill.

1 The candidate

After notebook 18 the four-variable law

$$\hat{s} \approx 0.14 \gamma_1^{1.32} (\gamma_2 - \gamma_1)^{0.45} e^{-0.13D} 1.32^{[\text{odd}]}$$

is dead on its documented fragility axes. The leftover proposal was S2 stripped to two variables: desert and parity, the two inversions already in the depth note ($\chi_4 > \chi_5$, $\chi_7 > \chi_8$).

Engine `code/map2.py`, training set = the thirteen stationary rows of [?] (uniformized at $\mu = 30\text{--}38$; χ_{15} dropped as rising; ζ kept out as a pole). Least squares in log s :

$$s = 0.171 \gamma_1^{1.327} 1.54^{[\text{odd}]}.$$

Training rms 21%, against 6% for the four-variable law.

2 The hold-out

χ	s (unif.)	$\hat{s}_2$	err	$\hat{s}_4$
χ_{-8}	1.30	1.42	+10%	1.47
χ_{-20}	0.68 [†]	0.82	+21%	0.57
χ_{-23}	0.54	1.06	+97%	0.76

[†] still rising at $\mu = 38$. χ_{-23} is stationary ($0.49 \rightarrow 0.53$). The two-variable map fails the same kill, worse.

3 Parity is the axis

χ_{-23} is odd, $\gamma_1 = 2.87$. Even characters at that desert: χ_{24}^e ($\gamma_1 = 2.69$, $s = 0.50$), χ_{21} (2.32, 0.58). Odd characters at that desert: χ_{11} (2.48, 1.07). Measured $s(\chi_{-23}) = 0.54$ sits on the *even* locus. The factor $1.54^{[\text{odd}]}$ is what inflates 1.06.

A three-variable refit with no parity, $\log s = a + b \log \gamma_1 + c \log(\text{gap}) + d \log D$, gives $\hat{s}(\chi_{-23}) = 0.45$ against 0.54 (−16%, inside the old 20% kill). Training rms 17%. This is a diagnosis, not a successor map: it was read on the same hold-out that killed the predecessor. The next map, if there is one, has to be preregistered before the next character.

4 Status

$s = s(\gamma_1, \text{parity})$ is false. S2 remains: s is $-d \log |\hat{v}_0(\gamma_1)|/d\mu$. The arithmetic of that Lyapunov exponent is not parity-times-desert. χ_{-23} is the witness.